

THE RATTY2TAILS PAPERS

MONETARY HISTORY SERIES



The Vault That Was Never There

*Fractional reserve banking, the money multiplier myth,
and what actually constrains a bank*

RATTY2TAILS

September 2026

TL;DR

- The story most people carry around — that banks lend out a fraction of the deposits they receive, and a reserve ratio caps how far the chain can run — is a teaching model, not a description of a working bank. The Bank of England said so in its own quarterly bulletin in 2014, and said it plainly.
- In practice the causation runs the other way. A bank does not need your deposit in order to lend. It writes a loan on one side of its ledger and a brand-new deposit on the other, and the money supply is larger by the time the ink dries. Reserves are acquired afterwards, if at all.
- The reserve requirement, the supposed brake, has been quietly dismantled almost everywhere. The United Kingdom abandoned it in 1981. Canada abolished it in 1992. The Federal Reserve cut it to zero in March 2020 and it has stayed there. The euro area's 1 per cent is the last significant survivor in the developed world.
- What actually binds a bank is capital, liquidity regulation, the cost of funding and the availability of creditworthy borrowers — not a vault ratio. The multiplier is a residual, not a constraint.
- Fractional reserve banking works by promising instant redemption on liabilities backed by assets that cannot be liquidated instantly. That mismatch is the entire business model, and also the entire fragility. Silicon Valley Bank lost roughly a quarter of its deposits in a single day in March 2023.
- The system is held together not by reserves but by scaffolding: deposit insurance (£120,000 in the UK since December 2025) and a central bank willing to lend against collateral at a penalty rate, as Bagehot prescribed in 1873.
- The alternatives — the Chicago Plan, Fisher's 100% Money, Switzerland's Vollgeld referendum — have been proposed repeatedly for ninety years and rejected every time, most recently by three-quarters of Swiss voters in June 2018.
- The euro area is currently debating doubling its reserve ratio to 2 per cent. The stated motive is not monetary control or depositor safety. It is to reduce the Eurosystem's own interest bill by roughly €4 billion a year.
- A bearer asset held in self-custody is the one instrument in this essay that has no counterparty, no reserve ratio and no scaffolding. That is its entire proposition, and its entire burden.

INTRODUCTION AND SCOPE**The Question Nobody Asks at the Cashpoint**

There is a particular kind of confidence that comes from standing at a cash machine. You put in a card, you ask for fifty pounds, and fifty pounds comes out. The transaction feels like retrieval. Somewhere, you assume, there is an account, and in that account there is money, and the machine has gone and fetched some of it.

The truth is stranger and more interesting. There is no pile. What you have is not money in the sense of a thing you own, but a promise — an entry in a private company's ledger recording that the company owes you fifty pounds on demand. The company has made a great many such promises, and it holds nowhere near enough cash to honour them all at once. It never has. It is

not designed to. If every customer of your bank asked for their balance in notes on the same Tuesday morning, the bank would fail before lunch, and it would fail not because it was badly run or dishonest but because that is what a bank is.

This arrangement has a name — fractional reserve banking — and it is the most consequential piece of financial plumbing in the modern world. It is also the most thoroughly misunderstood, and the misunderstanding is not confined to the general public. It is embedded in textbooks, repeated by journalists, and reproduced in a very large fraction of the commentary written by people who are certain they have seen through the system. A great deal of sound-money criticism attacks a model of banking that no longer describes any bank in any major economy, and possibly never did.

The purpose of this essay is to replace that model with the mechanics. We will begin with the origin story — the goldsmiths of seventeenth-century London — and note where it is illuminating and where it misleads. We will then set out the textbook money multiplier in full, because it needs to be understood before it can be dismantled, and then dismantle it using the central banks' own published descriptions of what they do. We will trace the quiet disappearance of the reserve requirement across the developed world, a regulatory abolition that happened in plain sight and attracted almost no comment. We will ask what actually restrains bank lending, since something evidently does. We will look at the fragility that the arrangement necessarily creates, at the scaffolding erected to contain that fragility, and at the distributional consequences of granting the power of money creation to a small number of private institutions. We will examine the full-reserve alternatives that have been proposed since 1933 and rejected every time. And we will close, as this series tends to, with the question of what a bearer asset does to any of it.

Two things this essay is not. It is not a claim that banking is a fraud; the legal and accounting structure is public, documented and, at least in its mechanics, honestly described by the institutions that operate it. And it is not a policy prescription. It is an attempt to establish what is actually true about how money comes into existence, on the reasonable premise that you cannot sensibly hold an opinion about a system you have mis-specified.

Whenever a bank makes a loan, it simultaneously creates a matching deposit in the borrower's bank account, thereby creating new money.

BANK OF ENGLAND QUARTERLY BULLETIN, 2014 Q1

That sentence was published by a central bank, in a house journal, under the names of three economists in its Monetary Analysis Directorate. It is not a fringe position. It is not heterodox. It is the operating description of the institution that issues sterling. And it is flatly incompatible with what most people believe banking to be.

SECTION I

The Goldsmith's Tale, and Why It Is Only Half True

Every account of fractional reserve banking begins in the same place: a goldsmith's shop on Lombard Street, some time after the Civil War. The story runs as follows. Merchants deposit gold with goldsmiths for safekeeping, because goldsmiths have strongrooms and merchants do not. The goldsmith issues a receipt. The receipts prove more convenient to trade than

the metal, and begin to circulate as money in their own right. The goldsmith, observing that the gold mostly sits still — that on any given day only a small fraction is redeemed — begins issuing receipts against gold that is not there, lending them at interest. He has discovered that he can create claims out of nothing so long as not everyone asks at once. Banking is born, and its original sin with it.

It is a marvellous story. It is compact, it has a villain, and it makes the essential point — that redemption promises can exceed the stock of the thing promised — with great economy. It is also, as history, unreliable in ways that matter.

The difficulty is with the framing of "safekeeping". Goldsmith-bankers in Restoration London were not principally custodians who then betrayed a trust. They were dealers in bullion and foreign exchange who moved, fairly quickly, into deposit-taking and note-issue as a commercial business. The arrangements were largely understood by the parties to them: depositors received interest, and you do not pay someone interest to guard their property in a box. Interest is the price of lending. The moment a depositor accepts a yield, he has stopped being a bailor and started being a creditor, and the gold is no longer his in any meaningful sense — he has swapped it for a claim. The Roman law distinction is between *depositum regulare*, where the identical thing must be returned, and *depositum irregulare* or *mutuum*, where an equivalent must be returned and title has passed. English commercial practice, and eventually English law, settled firmly on the latter for money.

The settling was not painless. It took until 1848 and the House of Lords decision in *Foley v Hill* to make the principle unambiguous, and the ruling is worth dwelling on because it is the legal foundation of everything that follows. Their Lordships held that money paid into a bank ceases, entirely, to be the money of the customer. It becomes the banker's money, to do with as he pleases. He is not a trustee and not an agent. He is a debtor, bound only to repay an equivalent sum when asked. The customer, who imagines he has money *in* a bank, in fact has no money at all: he has an unsecured claim on a leveraged private company.

This is not a technicality. It is the single most important fact about a bank account, and it is not taught, not disclosed at account opening, and not present in the mental model of nearly anyone who holds one. The cash machine feels like retrieval because the law's answer — that you are an unsecured creditor collecting on a debt — is not an answer anyone wants at eight in the morning.

WHAT THE GOLDSMITH STORY GETS RIGHT

Strip away the fraud framing and the mechanism survives intact. Redeemable claims circulate more conveniently than the underlying asset. Circulating claims are redeemed only fractionally on any given day. An issuer who observes this can issue more claims than he holds assets, and will earn a return for doing so. The incentive is structural and it does not require anyone to be wicked. That is the durable insight, and it is why the tale keeps getting told even though the history is shaky.

What the tale obscures is the direction of causation, and this is where it does real damage. It leaves the impression that a bank first acquires a reserve and then issues claims against it — that the deposit precedes the loan, that the gold in the strongroom is the raw material from which

lending is manufactured. Three and a half centuries later, that impression is still the default, and it is the thing that has to go.

SECTION II

The Textbook Machine

Before dismantling the money multiplier, it deserves a fair and complete statement, because it is elegant and because a great deal of monetary intuition is built on it.

Assume a central bank injects £1,000 of new reserves into the banking system — say, by buying a bond from a member of the public, who deposits the proceeds. Assume a reserve requirement of 10 per cent: every bank must hold reserves equal to at least a tenth of its deposits. Assume, finally, that banks hold no excess reserves and that borrowers redeposit everything they receive.

Bank A now holds a £1,000 deposit and £1,000 of reserves. It must retain £100; the other £900 is surplus, and surplus reserves earn little or nothing, so it lends them. The borrower spends the £900, and the recipient banks it at Bank B. Bank B now has a £900 deposit, retains £90, lends £810. Bank C receives £810, retains £81, lends £729. And so on, each round smaller than the last by a factor of 0.9.

The sum of the series is the initial injection divided by the reserve ratio: $£1,000 \div 0.1 = £10,000$. One thousand pounds of central bank money supports ten thousand pounds of deposits. The reserve ratio is the multiplier's reciprocal, and therefore the central bank's control lever: tighten the ratio and the pyramid shrinks; loosen it and the pyramid grows. Money creation becomes a matter of arithmetic, and monetary policy a matter of dialling in a number.

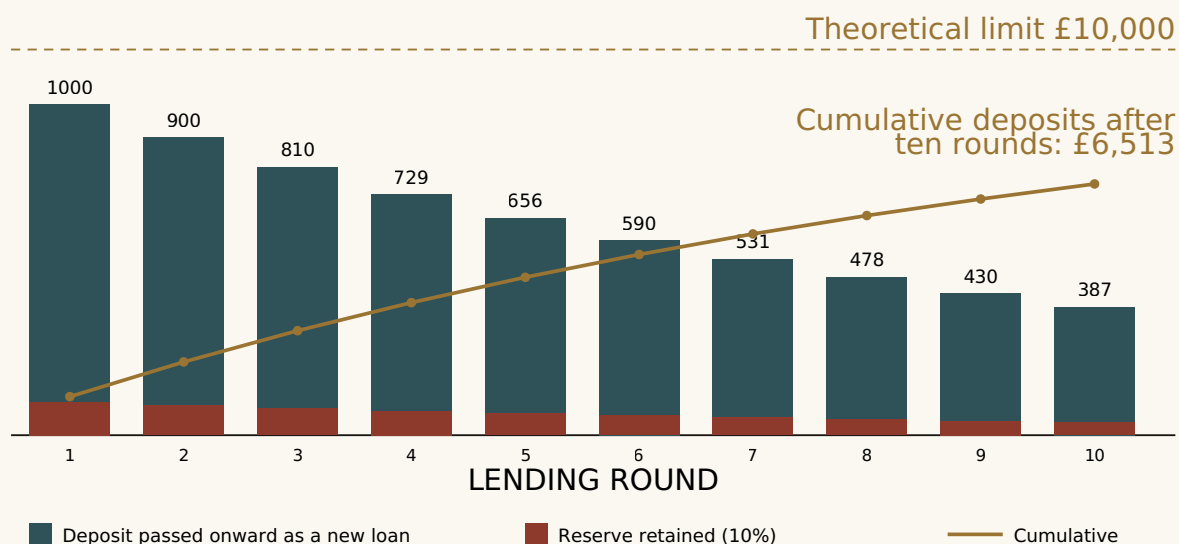


FIGURE 1 The textbook cascade. An injection of £1,000 into a system with a 10 per cent reserve requirement, traced through ten rounds of lending and redepositing. Each round is 90 per cent of the one before; the sliver retained as reserve shrinks with it. The series converges on £10,000, but slowly — after ten rounds it has reached only £6,513, and it takes forty-four rounds to get within one per cent of the limit. The convergence is asymptotic, which is the first hint that the "limit" is doing less work than it appears to.

The model is taught because it is teachable. It explains the difference between base money and broad money in one diagram. It gives the central bank a clean lever. It has an intuitive moral: money is created by banks, out of something, in a chain, up to a ceiling set by the regulator.

Every element of that moral is wrong, except the first.

Where the model breaks

The assumptions are the giveaway. The model requires that banks hold no excess reserves — but banks in the United States, the United Kingdom and the euro area have held enormous excess reserves continuously since 2009, and the pyramid did not grow to fill them. It requires that reserves be scarce and therefore binding — but under an ample-reserves operating framework they are deliberately abundant, and abundance is the opposite of a constraint. It requires that the central bank control the quantity of reserves — but a modern central bank targets a *price*, an overnight interest rate, and supplies whatever quantity of reserves is consistent with that price. It cannot simultaneously fix the quantity and the price, and it has chosen the price.

Most fundamentally, the model has the sequence backwards. It has a bank waiting for a deposit before it can lend. No loan officer in any commercial bank has ever checked the reserve account before approving a mortgage.

SECTION III

What Banks Actually Do

In the spring of 2014, the Bank of England published a paper in its *Quarterly Bulletin* called *Money creation in the modern economy*. It was written by Michael McLeay, Amar Radia and Ryland Thomas of the Monetary Analysis Directorate, and it did something central banks rarely do: it stated, in unhedged prose, that the model in the textbooks was wrong, and then explained why.

The paper's argument is simple. A bank does not act as an intermediary passing saver's money to borrowers. When it makes a loan, it creates the deposit at the same instant, by the same act. The loan is an asset for the bank — a claim on the borrower. The deposit is a liability — a claim against the bank. Both appear simultaneously, both are entries in the bank's own ledger, and the second of them is money. Ninety-seven per cent of the money in circulation in the United Kingdom exists in this form: as bank deposits, which is to say as bank liabilities, which is to say as the by-product of lending.

Consider what this means for a mortgage. You are approved for £200,000. The bank does not go and find £200,000 that somebody else saved. It credits the seller's account with £200,000 that did not exist a moment earlier, and books a £200,000 loan against your name. Its balance sheet is now £200,000 larger on both sides. The broad money supply has grown by £200,000. No reserves moved. No saver was intermediated. No vault was opened.

THE MOMENT OF CREATION

Before the loan

ASSETS		LIABILITIES	
Reserves	50	Deposits	890
Loans	800	Debt	50
Other assets	150	Equity	60
Total	1,000	Total	1,000

Reserves unchanged — nothing was lent out of anybody else's savings.

After the loan

ASSETS		LIABILITIES	
Reserves	50	Deposits	1,090
Loans	1,000	Debt	50
Other assets	150	Equity	60
Total	1,200	Total	1,200

New deposit = new money, created by the act of lending itself.

£200
loan

FIGURE 2 A £200 loan on a stylised bank balance sheet, in units of £ million. The balance sheet grows on both sides simultaneously. Loans rise by 200 because the bank has acquired a claim on the borrower; deposits rise by 200 because the bank has issued a claim against itself. Reserves do not move, and equity does not move. The bank is more leveraged than it was, and the broad money supply is larger. This is money creation, and it is an accounting operation, not a transfer.

The reserves question does not vanish; it is merely deferred. Once the seller spends the new deposit, and the recipient banks elsewhere, our bank must settle with the receiving bank in central bank reserves. If it lacks them, it borrows them — in the overnight interbank market, or from the central bank itself at the standing facility. Reserves are a settlement asset, acquired *after* lending, at a price the central bank sets. They are not raw material acquired before.

This is the crux. If reserves are supplied on demand at a policy rate, then the reserve ratio cannot be a quantitative limit. It is a cost, and a small one. The central bank does not ration the quantity of money; it prices credit and lets the quantity fall out. The multiplier, in the Bank of England's own framing, is a ratio observed after the fact rather than a mechanism that governs anything.

THE HERESY THAT WASN'T

This account is sometimes presented as a radical revelation, particularly by people selling something. It is not. Knut Wicksell described credit-driven money creation in 1898. Joseph Schumpeter set out the "credit theory of money" against the "money theory of credit" in 1954 and was scathing about the latter. Basil Moore's *Horizontalists and Verticalists* made the endogenous-money case at book length in 1988. The Deutsche Bundesbank published its own version of the Bank of England's explanation in April 2017. The 2014 bulletin was notable not for its content but for its author: it is one thing for a heterodox economist to say the textbook is wrong, and quite another for the institution that prints the currency.

None of which means banks can lend without limit. They plainly cannot; if they could, every bank would be infinitely large. Something binds. It is simply not the thing everyone thinks binds — and to see what it is, it helps first to watch the supposed constraint disappear.

SECTION IV

The Quiet Abolition

If the reserve requirement were the load-bearing constraint on money creation, its removal would be a constitutional event. Parliaments would debate it. Newspapers would cover it. Something, one imagines, would happen.

It was removed nearly everywhere, and almost nothing happened.

The United Kingdom went first among the large economies. The 1.5 per cent reserve asset ratio was abolished in 1981 as part of the Thatcher government's monetary control reforms, and sterling has had no reserve requirement since. What remains — cash ratio deposits, non-interest-bearing balances that banks lodged with the Bank of England — was never a monetary instrument at all. It was a funding arrangement: a levy dressed as a ratio, designed to pay for the Bank's policy and supervisory functions out of the forgone interest.

Canada followed, phasing its requirement out under the 1991 Bank Act revisions and reaching zero by June 1994. New Zealand, Australia, Sweden, Switzerland and Hong Kong all abandoned theirs across the same period. By the mid-1990s a substantial share of the developed world was operating banking systems with no legal reserve ratio whatsoever, and the sky remained where it was.

The United States held on longest, and its capitulation is the most instructive. On 15 March 2020, with the pandemic beginning, the Federal Reserve Board announced that reserve requirement ratios would be reduced to zero, effective 26 March. The action eliminated reserve requirements for all depository institutions in the United States and released an estimated \$200 billion of previously required reserves. The stated rationale was not emergency relief. It was that the Fed had moved to an ample-reserves operating framework, in which the federal funds rate is steered by the interest paid on reserve balances rather than by making reserves scarce. Reserve requirements had simply stopped doing anything.

THE DISAPPEARING RESERVE REQUIREMENT

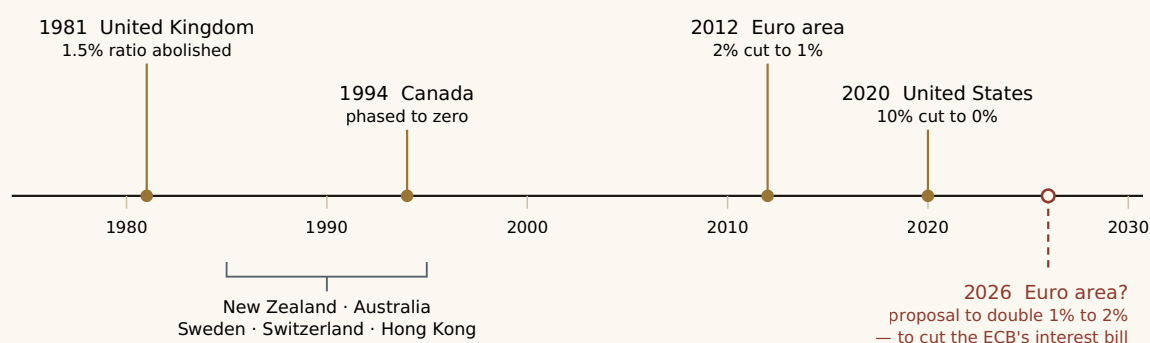


FIGURE 3 Four decades of abolition. The supposed brake on money creation was released, jurisdiction by jurisdiction, across the developed world, with no discernible discontinuity in credit growth attributable to the change. The euro area's surviving 1 per cent is now the subject of a proposal to double it — for reasons, as Section IV explains, that have nothing to do with restraining lending.

RESERVE REQUIREMENTS IN SELECTED JURISDICTIONS, 2026

JURISDICTION	REQUIREMENT	NOTES
United Kingdom	None	Abolished 1981. Banks hold reserves voluntarily under a remunerated framework.
United States	Zero	Cut from 3% / 10% to zero on 26 March 2020; unchanged since.
Canada	None	Phased out under the 1991 Bank Act; fully zero by June 1994.
Euro area	1%	On short-maturity liabilities. Remunerated at 0% since September 2023. Doubling under discussion.
Australia, New Zealand, Sweden, Switzerland, Hong Kong	None	Abolished during the deregulation of the 1980s and 1990s.
China	Active tool	The PBoC adjusts its reserve requirement ratio several times a year, with separate ratios by bank size.
India	Active tool	Cash reserve ratio used alongside the statutory liquidity ratio.

There is a detail in the American arrangement that deserves to be framed and hung on a wall. The Federal Reserve Act still obliges the Board to index two numbers each year: the reserve requirement exemption amount and the low reserve tranche. For 2026 these were duly set at \$39.2 million and \$674.1 million respectively, published in the Federal Register in November 2025, and accompanied by a note observing that the indexation is required by statute but will not affect depository institutions' reserve requirements, which remain zero. Every year, the central bank of the United States calculates, to one decimal place, the thresholds at which a rate of zero begins to apply to a different quantity of nothing.

The euro area exception, and what it reveals

The euro area is the last major holdout. Credit institutions must hold 1 per cent of specified short-maturity liabilities — mainly customer deposits and debt securities with maturities under two years — as reserves at their national central bank, averaged over a maintenance period of six or seven weeks. The ratio was set at 2 per cent in 1999 and halved to 1 per cent in January 2012, at the depth of the sovereign debt crisis. Since September 2023 those required reserves have been remunerated at zero.

In June 2026 it emerged that the Governing Council was considering doubling the ratio back to 2 per cent. Reuters reported the story from Sintra at the end of that month; Christine Lagarde confirmed at the July press conference that the subject had not been discussed at that meeting but would be. A decision was expected in the autumn. As this essay goes out, the ratio is still 1 per cent and the matter is unresolved.

What makes the proposal remarkable is the stated motive. It is not that credit is growing too fast. It is not that banks are insufficiently liquid. It is that the Eurosystem pays interest on excess reserves, that interest is expensive, and shifting a slice of those balances into an unremunerated required-reserve bucket would save the ECB and the nineteen other national central banks something on the order of €4 billion a year. Doubling the ratio would drain perhaps €165 billion

of remunerated liquidity. The Association of German Banks objected, and objected on the merits — that it would tie up liquidity, weaken profitability and reduce the scope for lending.

Article 19 of the Statute of the European System of Central Banks permits the ECB to require minimum reserves *in pursuance of monetary policy objectives*. Reducing the central bank's own interest expense is a great many things, but it is difficult to describe it as a monetary policy objective. The reserve requirement, having been emptied of its monetary function, is being repurposed as a seigniorage-recovery device: an unremunerated levy on bank balance sheets, defended on the grounds that the levying institution would like the money.

That is the clearest evidence available that the reserve ratio no longer constrains lending. If it did, no central bank would propose doubling it as an accounting convenience.

SECTION V

What Actually Binds

Something restrains bank lending, because banks are finite. Four things do, and none of them is a vault ratio.

Capital

The binding regulatory constraint in every developed banking system is equity, not reserves, and the distinction is worth being precise about because the two are routinely confused. Reserves are an *asset*: a balance at the central bank, sitting on the left of the balance sheet. Capital is the gap between assets and liabilities — the owners' stake, sitting on the right. A reserve requirement says *hold this much of your assets in this particular form*. A capital requirement says *fund this much of your assets with money that cannot run*.

Under Basel III as implemented across the G20, a bank must hold common equity Tier 1 capital of at least 4.5 per cent of risk-weighted assets, plus a capital conservation buffer of 2.5 per cent, plus countercyclical and systemic buffers, plus whatever its supervisor adds under Pillar 2. A large European or British bank typically operates against a requirement in the region of 11 to 14 per cent, and holds a management buffer above it. There is also a non-risk-weighted backstop — the leverage ratio — precisely because risk weights are a model output and models can be gamed.

Every new loan consumes capital, because it adds risk-weighted assets. A bank that wants to lend more must either raise equity, retain earnings, or shed assets elsewhere. That is a real, hard, immediate constraint, and it is the one that actually governs the size of bank balance sheets. When the Bank of England's Financial Policy Committee wishes to lean against a credit boom, it does not adjust a reserve ratio. It adjusts the countercyclical capital buffer.

Liquidity

Basel III also imposes two liquidity standards. The Liquidity Coverage Ratio requires a bank to hold enough high-quality liquid assets to survive thirty days of a stipulated stress scenario. The Net Stable Funding Ratio requires longer-term assets to be matched with longer-term funding. These are the modern descendants of the reserve requirement, and they are considerably more sophisticated: they distinguish between a sticky insured retail deposit and a hot uninsured cor-

porate balance, and they weight the assets by how fast they can actually be sold without a haircut.

Cost

Reserves are not free. A bank that lends aggressively will lose deposits to other banks as borrowers spend, and must replace that funding — from the interbank market, from wholesale issuance, or from the central bank at the standing facility. The policy rate sets that cost. This is the channel through which monetary policy actually transmits: not by rationing reserves, but by making the marginal loan more or less profitable.

Borrowers

The most under-rated constraint of all is that a loan requires someone willing to take it and creditworthy enough to be given it. Banks in the euro area and Japan have spent much of the last fifteen years with vast excess reserves and inadequate demand for credit at rates that compensate for risk. You cannot push a balance sheet with a string.

WHICH CONSTRAINT BITES FIRST?

Schematic room-for-manoevre left by each constraint; the shortest bar is the one that binds.

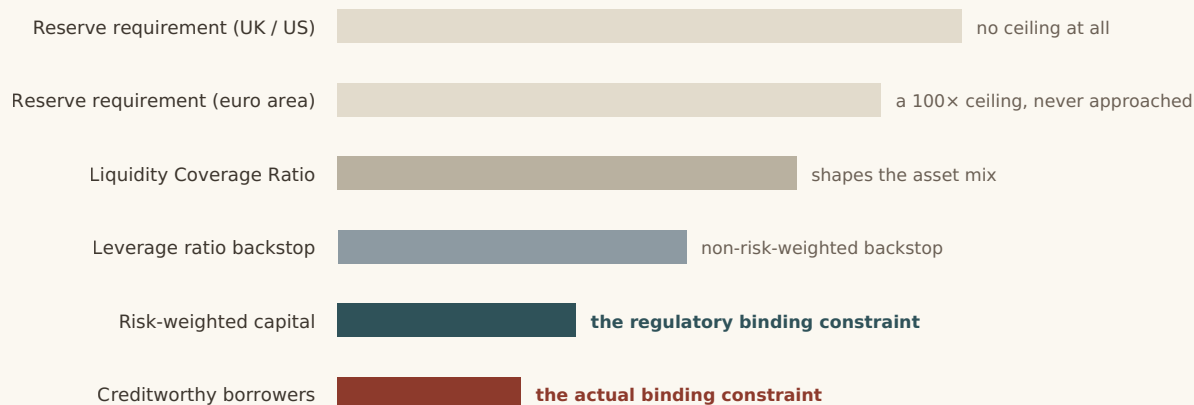


FIGURE 4 A schematic ordering, not a measurement. The point is the ranking, not the lengths: the reserve requirement, where it survives at all, leaves so much room that it is never the operative limit, while capital adequacy and the supply of creditworthy borrowers do the real work. A constraint that never binds is not a constraint. It is a decoration.

Notice what this reordering does to the popular critique. The objection that banks "lend money they don't have" is, in the strict sense, correct — but it is correct in a way that has nothing to do with reserve ratios, and fixing the ratio would not address it. The interesting question is not *how thin is the reserve?* It is *who is allowed to issue the liabilities that the rest of us treat as money, and on what terms?*

SECTION VI

The Mismatch, and Its Consequences

A bank's liabilities are short, certain and demandable. Its assets are long, uncertain and illiquid. That sentence contains the whole business model and the whole fragility, and the two

cannot be separated, because the profit is the risk. Banks earn the spread between long rates and short rates, and the spread exists precisely because bearing that mismatch is dangerous.

MATURITY TRANSFORMATION: THE SHAPE OF A BANK

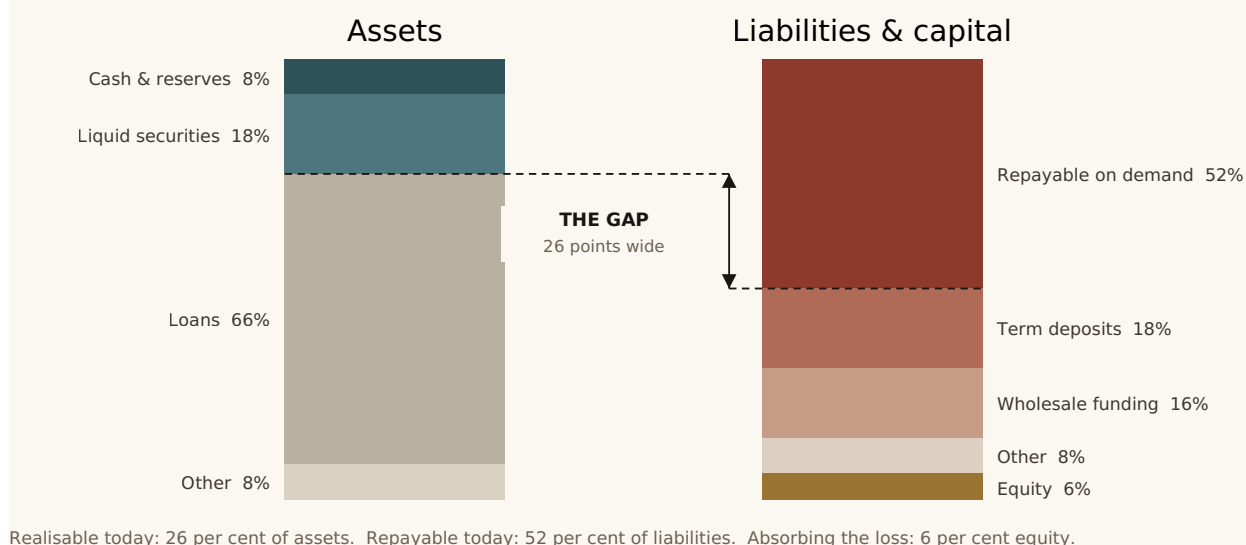


FIGURE 5 A stylised commercial bank, with segments ordered by liquidity on the left and by how quickly the money can leave on the right. The bank can meet roughly a quarter of its obligations out of assets it can realise at once, against liabilities of which more than half are payable on demand. That gap is not a flaw in the design. It is the design, and it earns the spread that pays for the branch network.

Douglas Diamond and Philip Dybvig formalised the consequence in 1983, and were given the Nobel Prize for it alongside Ben Bernanke in 2022. Their model has two equilibria. In the good one, depositors who do not need their money leave it alone, the bank holds its long assets to maturity, and everybody is paid. In the bad one, each depositor expects the others to withdraw, and since the bank pays out in the order people arrive, the rational response to expecting a run is to run. Both equilibria are self-fulfilling. Nothing about the bank's assets need change between them. The trigger can be a rumour, an accounting disclosure, or a group chat.

Britain learned this in September 2007, when queues formed outside Northern Rock — the first visible run on a British bank since Overend, Gurney collapsed in 1866. Northern Rock's problem was not fraud but funding structure: it had grown fast on wholesale money and securitisation, and when those markets shut in August 2007, it could not roll its liabilities. The retail queues were a consequence of the wholesale run, not its cause.

What happened at Silicon Valley Bank sixteen years later showed how much faster the mechanism now runs. SVB had taken in enormous deposits during the 2021 venture boom and parked them in long-dated Treasuries and mortgage-backed securities. When rates rose sharply in 2022, those securities lost market value. The accounting permitted the bank to carry much of the portfolio at amortised cost, so the losses were disclosed but not recognised. On 8 March 2023 the bank announced a \$1.8 billion loss on securities sales and a plan to raise capital. The disclosure was the trigger.

On 9 March, depositors and investors attempted to withdraw \$42 billion — roughly a quarter of the bank's deposits — in a single day. By the close of business SVB had a negative cash balance of \$958 million and could not meet its cash letter with the Federal Reserve. The following morning, the bank informed regulators that a further \$100 billion was scheduled to leave. Michael Barr,

the Fed's Vice Chair for Supervision, later told the Senate Banking Committee what those two figures meant together: \$142 billion, or 81 per cent of a \$175 billion deposit base, over two days. Regulators closed the bank on 10 March.

WHY THE RUN WAS FASTER THAN ANY BEFORE IT

Three things had changed since 1866, and since 2007. Deposits could be moved by phone, at any hour, without a queue or a conversation. SVB's depositors were unusually concentrated and unusually connected — a single industry, largely advised by the same handful of venture firms, many of whom told their portfolio companies to withdraw at the same moment. And the great majority of those balances were above the insured limit, so there was no reason not to run. Diamond and Dybvig's bad equilibrium used to take days to assemble. It now assembles in an afternoon.

The lesson is not that SVB was a badly run bank, though it was. It is that the mismatch is unavoidable in any institution that promises instant redemption while holding assets that cannot be instantly sold, and that the speed at which the promise can be tested has increased by orders of magnitude while the structure of the promise has not changed at all.

SECTION VII

The Scaffolding

If the mismatch is unavoidable and the run equilibrium is always available, why is there not a bank failure every month? Because the mismatch is not, in fact, borne by the banks. It has been socialised, deliberately and openly, by two pieces of public architecture.

Deposit insurance

The first is a government promise that the small depositor will be repaid regardless. In the United Kingdom, the Financial Services Compensation Scheme protects deposits per person, per authorised institution. The limit had stood at £85,000 since 2017; the Prudential Regulation Authority consulted in March 2025 on raising it to £110,000, settled on £120,000 in light of the responses and the inflation data, and the new limit took effect for firm failures on or after 1 December 2025. The cap on temporary high balances — the money that sits in your account for a few months after you sell a house — rose at the same time from £1 million to £1.4 million. In the United States the FDIC limit is \$250,000; in the euro area, €100,000.

Deposit insurance does not make banks safe. It makes the run *pointless* for anyone under the limit, which is a different thing and, for the purposes of financial stability, a better one. If you know you will be repaid, you have no reason to queue, and if nobody queues the bad equilibrium never assembles. The scheme works by removing the incentive to test the promise.

It works less well for those above the limit, which is exactly what SVB demonstrated: the overwhelming majority of its deposits were uninsured, and uninsured money has every reason to move first. The American authorities' response was to invoke the systemic risk exception and guarantee *all* deposits at SVB and Signature Bank, insured or not. The \$250,000 limit turned out, in the moment, to be advisory. Everyone in the market drew the appropriate inference.

The lender of last resort

The second piece of scaffolding is older, and was described with such clarity in 1873 that we still cite the description. Walter Bagehot's *Lombard Street* set out what a central bank should do in a panic: lend freely, at a penalty rate, against good collateral, to solvent institutions. Each clause does work. *Freely*, because half-measures invite the panic to continue. *At a penalty rate*, so that the facility is used in emergency rather than in the ordinary course of business. *Against good collateral*, so that the central bank is providing liquidity, not capital. *To solvent institutions*, because an insolvent bank should be resolved, not funded.

The doctrine has been honoured with increasing looseness. In March 2023 the Federal Reserve established the Bank Term Funding Program, which lent against Treasury and agency collateral valued *at par* — that is, at face value, rather than at the market price that had fallen and caused the problem in the first place. The programme was a sensible piece of firefighting and it worked. It was also, unmistakably, a departure from the third clause: collateral valued at what it is worth is good collateral; collateral valued at what one wishes it were worth is a subsidy.

THE BARGAIN, STATED PLAINLY

Private institutions are permitted to create the money supply as a by-product of profitable lending, and to keep the profits from doing so. The public bears the tail risk through deposit insurance and central bank liquidity. In exchange, the institutions accept capital requirements, liquidity requirements, supervision and resolution planning. This is not a conspiracy; it is a published, negotiated, endlessly revised settlement. Whether it is a *good* settlement is a separate question, and a fair one. But it should be argued about as what it is, rather than as a secret.

SECTION VIII

Who Gets It First

The mechanics matter for a reason beyond accuracy, and it is the reason this series keeps returning to Richard Cantillon. If new money enters the economy at a point, rather than everywhere at once, then the people standing near that point receive it at the old prices and everyone else receives it, if at all, at the new ones. The distributional consequence is not a side effect of monetary expansion. It is the first effect.

Under fractional reserve banking, the point of entry is a lending decision made by a commercial bank. That decision is made on commercial criteria, and the criteria reward collateral. Land and buildings are excellent collateral: they are hard to move, easy to value, and can be repossessed. A software business with a good idea and no assets is poor collateral. The predictable result — visible in the balance sheets of every developed banking system — is that the overwhelming share of new bank credit is secured on property rather than extended to productive enterprise.

This is a self-reinforcing loop, and it is the most important consequence of the whole arrangement. Bank lending bids up house prices. Higher house prices make housing better collateral. Better collateral supports more lending. Wages do not participate in this loop, because wages are not collateral. Anyone who already owns the asset gains; anyone trying to buy it faces a price set

by the availability of credit rather than by the cost of construction. Britain has run this experiment for forty years with unusual thoroughness, and then expressed bafflement at the result.

None of this requires malice, and none of it is well described as a conspiracy. It is what you get when the power to create money is delegated to institutions whose commercial interest is to lend against whatever is easiest to repossess. The system is doing exactly what its incentives specify.

SECTION IX

The Alternatives, and Why They Keep Losing

The proposal to abolish fractional reserve banking is almost a century old and has never been close to enactment in any major economy. It is worth understanding why, because the reasons are more interesting than either side usually admits.

The Chicago Plan

In March 1933, in the wreckage of the American banking system, a group of economists at the University of Chicago — Henry Simons, Frank Knight, Paul Douglas and others — circulated a six-page memorandum proposing that deposit banking be separated from lending entirely. Demand deposits would be backed one-for-one by government-issued money. Banks could still lend, but only funds actually entrusted to them for a term, in what would today be called a mutual fund. Runs would become logically impossible, because there would be nothing to run on.

Irving Fisher took up the proposal and published *100% Money* in 1935, arguing it would eliminate bank runs, drastically shrink the business cycle and allow the national debt to be retired. Fisher lobbied Roosevelt directly. He lost to a far more modest reform — federal deposit insurance — which achieved the immediate objective of stopping the runs without disturbing the banks' business model at all. That trade, safety through a public guarantee rather than through a structural change, is the trade every subsequent generation has also made.

The plan has been revived repeatedly. Jaromír Beneš and Michael Kumhof published *The Chicago Plan Revisited* as an IMF working paper in 2012, modelling Fisher's claims in a modern framework and finding, to some surprise, that they broadly held up. Iceland's parliament commissioned a report on monetary reform along the same lines in 2015. Neither led anywhere.

Vollgeld

The only time an electorate has been asked directly was in Switzerland on 10 June 2018. The *Vollgeld* initiative would have made the Swiss National Bank the sole creator of Swiss francs, converting commercial bank deposits into central bank money and confining banks to lending funds actually entrusted to them. The proposal reached a national referendum, was opposed by the SNB, the government, the banks and almost every economist who commented on it, and was rejected by roughly three-quarters of those who voted.

The objections were not primarily ideological. Credit allocation would shift towards whoever administered the new system. A country that unilaterally adopted full reserve would watch credit intermediation migrate offshore, or into whatever shadow-banking structure grew to replace it — a point worth sitting with, because it suggests that maturity transformation is not a

policy choice at all but an economic function that will be performed by *something*, regulated or not. And the transition, involving the conversion of the entire deposit base, is a genuinely frightening operation to attempt in a live economy.

The free-banking objection, from the other direction

It would be dishonest to present the sound-money tradition as unanimous here, because it emphatically is not. Murray Rothbard and Jesús Huerta de Soto argue that fractional reserve banking is a legal category error — that a demand deposit is a bailment, that issuing claims exceeding the goods bailed is fraud whatever the contract says, and that the business cycle follows from the resulting fiduciary media.

George Selgin and Lawrence White, working in the same broad tradition, disagree on nearly every point. Their argument is that the arrangement is contractual and consensual; that *Foley v Hill* got the law right because a deposit paying interest is self-evidently a loan; and that the historical record of banking systems without a central bank or deposit insurance — Scotland between 1716 and 1845 is the favourite example, along with Canada before 1935 — shows fractional reserve note issue disciplined effectively by clearing-house settlement and unlimited shareholder liability. On this view the pathology is not the fraction. It is the central bank, the deposit guarantee and the implicit bailout, which together remove the discipline that made the fraction safe.

These two positions are frequently conflated by people who have read neither. They lead to opposite policy conclusions: one to a legal prohibition enforced by the state, the other to the removal of the state from banking altogether.

The narrow-banking door that is already ajar

There is one route to full-reserve deposits that requires no legislation, and central banks have been building it for a decade: a retail central bank digital currency is a full-reserve deposit account. A digital pound held directly at the Bank of England would be a claim on the central bank, not on a leveraged commercial lender. It could not fail, and there would be nothing to run to.

Which is precisely why every design published so far has holding limits attached. The Bank of England's own consultation floated a cap in the region of £10,000 to £20,000 per person, and the stated reason was to prevent large-scale migration of deposits out of the commercial banking system. Read that again. The institution designing the system has explicitly capped the safe option in order to protect the funding model of the unsafe one. It is the frankest admission available that fractional reserve banking is not something the public would choose on the merits if a free alternative were placed beside it.

SECTION X

The Bearer Asset

On 30 December 2010, a little over a year after the genesis block, Hal Finney posted to the Bitcoin forum a short note about banking. His argument was that Bitcoin as designed could not possibly carry the world's transaction volume on-chain, and that the natural consequence would be Bitcoin-backed banks issuing their own digital currencies redeemable in

bitcoin — with most transactions settling between banks rather than on the ledger. Bitcoin, in his framing, would function as the high-powered money of such a system, the reserve layer rather than the payment layer. He was, characteristically, unromantic about it: this was not a betrayal of the design but the expected outcome of it.

The forecast has aged uncomfortably well. Every major failure in the digital-asset industry has been a fractional reserve failure, and none of them required any new financial invention. Mt. Gox lost customer coins over a period of years while continuing to display balances. Celsius took customer deposits, promised yield, lent against them and could not meet redemption. BlockFi and Voyager did versions of the same. FTX moved customer assets to an affiliate and recorded them as though they had not moved. In each case the mechanism was ancient: issue more claims than you hold assets, rely on the fact that redemptions are fractional on any given day, and discover what Diamond and Dybvig proved — that the day they are not fractional is the day it ends.

This is the point at which the argument that "Bitcoin fixes this" needs to be handled carefully, because in its crude form it is false. Bitcoin does not prevent fractional reserve banking. Nothing prevents fractional reserve banking. Any asset that people would rather hold a convenient claim on than carry themselves will attract intermediaries who issue more claims than they hold, because the arbitrage is free and the discipline is deferred. The claim was true of gold in 1660 and it is true of bitcoin now. This series examined the same structure from the stablecoin side in *The Claim and the Coin*, and the shape was identical: a token trading at par with a dollar it may or may not be fully backed by, whose backing is a matter of published attestation rather than public verification.

What a bearer asset on a public ledger does change is narrower, and worth stating precisely.

First, **the reserve side becomes auditable**. A custodian can demonstrate control of specific outputs at a specific block height, cryptographically, without trusting an auditor. No goldsmith could do this and no bank can do it today. That is a genuine advance, and it is also a partial one — proof of reserves without proof of liabilities is theatre, because an institution that can show a large balance can still have issued claims against it many times over. The hard half of the problem is the liability side, and the cryptographic tools for it exist but are rarely deployed.

Second, and far more importantly, **there is an exit**. This is the part with no historical precedent. Under a gold standard, the ordinary person could in principle redeem notes for metal, but metal is heavy, hard to verify, hard to secure and impossible to move across a border in any quantity. The practical alternative to a bank was a mattress, and a mattress is not a monetary system. The option to opt out of the banking system existed on paper and dissolved on contact with logistics.

A twenty-four word seed phrase is different in kind. It is a bearer instrument with no counterparty, no reserve ratio, no deposit insurance and no lender of last resort, which can be held in a quantity limited only by what you can afford and moved anywhere at the speed of the network. It does not participate in the loop described in Section VIII, because no institution's lending decision expands its supply. Its issuance schedule was fixed in 2009 and is enforced by every node that validates a block.

That is the proposition. The burden that comes with it is exactly proportionate, and should not be soft-pedalled. Everything the scaffolding in Section VII does for you, it stops doing. There is no FSCS, no £120,000, no systemic risk exception, no Bagehot facility, no reversal of a mistaken

transaction and nobody to telephone. The failure mode of self-custody is not a bank run; it is a fire, a forgotten passphrase, a badly generated key or a bad afternoon. The system that stops working for you is also the system that stops working on you.

Foley v Hill settled in 1848 that money paid into a bank becomes the bank's money, and the customer becomes an unsecured creditor. A hundred and seventy years later, an industry with no knowledge of the case arrived at the same doctrine independently and compressed it into four words: not your keys, not your coins. It is the same sentence. It took the law a century and a half of commercial litigation to reach it, and it took the digital-asset industry about eight years of losing other people's money.

SUMMARY

What Is Actually True

The mental model most people hold — that banks lend out a fraction of the deposits they receive, and that a reserve ratio caps the chain — is not a simplification of the truth. It is a different claim, and it is wrong in its direction of causation, wrong about the constraint and wrong about the lever.

Banks create deposits by lending. The deposit is the loan, seen from the other side of the ledger, and ninety-seven per cent of the money in circulation exists in that form. Reserves are a settlement asset acquired afterwards at a price the central bank sets, not a raw material gathered beforehand. The reserve requirement, which was supposed to cap the process, has been abolished in the United Kingdom, Canada, Australia, New Zealand, Sweden, Switzerland and Hong Kong, and reduced to zero in the United States. The euro area's surviving 1 per cent is presently the subject of a proposal to double it in order to reduce the Eurosystem's own interest bill — which tells you everything about what the instrument is now for.

What binds a bank is capital, liquidity regulation, the cost of funding and the supply of borrowers who can plausibly repay. What makes a bank fragile is not the thinness of its reserve but the structure of its promise: instant redemption on liabilities backed by assets that cannot be instantly sold. That fragility is not a defect to be engineered away. It is the same thing as the profit, and it has been contained not by fixing it but by insuring it — through deposit guarantees and a central bank that will lend against collateral when the market will not.

The arrangement has consequences that are not accidental. Money enters the economy through a lending decision, lending decisions favour collateral, collateral means property, and the loop between credit and land prices runs unchecked while wages stand outside it. The alternatives have been on the table since 1933 and have lost every time — to deposit insurance in 1935, to the Swiss electorate in 2018, and to a holding limit in every CBDC design published since.

And a bearer asset held in self-custody does not abolish any of it. Intermediaries will re-emerge, because they always do, and some of them will issue more claims than they hold, because that is where the money is. What it offers is narrower and, in the long run, probably more useful: a reserve layer that can be audited rather than attested, and the first genuine exit in the

history of banking — available to anyone, at any size, without permission, and without a mattress.

Whether that exit is worth taking is a question each person answers for themselves. But it cannot be answered sensibly by anyone still holding the wrong model of what they would be leaving.

. . . .

REFERENCES & FURTHER READING

Sources

1. McLeay, M., Radia, A. and Thomas, R., *Money creation in the modern economy*, Bank of England Quarterly Bulletin, 2014 Q1.
2. McLeay, M., Radia, A. and Thomas, R., *Money in the modern economy: an introduction*, Bank of England Quarterly Bulletin, 2014 Q1.
3. Deutsche Bundesbank, *How money is created*, Monthly Report, April 2017.
4. *Foley v Hill* (1848) 2 HLC 28, House of Lords.
5. Bagehot, W., *Lombard Street: A Description of the Money Market*, London, 1873.
6. Diamond, D. W. and Dybvig, P. H., *Bank Runs, Deposit Insurance, and Liquidity*, Journal of Political Economy, vol. 91 no. 3, 1983.
7. Fisher, I., *100% Money*, Adelphi, New York, 1935.
8. Beneš, J. and Kumhof, M., *The Chicago Plan Revisited*, IMF Working Paper WP/12/202, 2012.
9. Huerta de Soto, J., *Money, Bank Credit and Economic Cycles*, Ludwig von Mises Institute, 2006.
10. Rothbard, M. N., *The Mystery of Banking*, Richardson & Snyder, 1983.
11. Selgin, G. and White, L. H., *In Defense of Fiduciary Media*, Review of Austrian Economics, vol. 9 no. 2, 1996.
12. Moore, B. J., *Horizontalists and Verticalists: The Macroeconomics of Credit Money*, Cambridge University Press, 1988.
13. Schumpeter, J. A., *History of Economic Analysis*, Oxford University Press, 1954.
14. Federal Reserve Board, *Reserve Requirements*, and press release of 15 March 2020 reducing reserve requirement ratios to zero effective 26 March 2020.
15. Federal Register, *Regulation D: Reserve Requirements of Depository Institutions*, 24 November 2025 — 2026 indexation of the exemption amount (\$39.2m) and low reserve tranche (\$674.1m).
16. European Central Bank, *Minimum reserves*, and press release of 27 July 2023 setting the remuneration of minimum reserves at 0%.
17. Statute of the European System of Central Banks and of the European Central Bank, Article 19.
18. Reuters, *ECB considers lifting banks' minimum reserves to lessen own losses*, 30 June 2026; and ECB press conference remarks, 23 July 2026.
19. Prudential Regulation Authority, PS24/25, *Depositor protection*, November 2025; Bank of England news release, 18 November 2025.
20. Barr, M. S., testimony before the Senate Committee on Banking, Housing, and Urban Affairs, 28 March 2023.
21. California Department of Financial Protection and Innovation, order taking possession of Silicon Valley Bank, 10 March 2023.
22. House of Commons Treasury Committee, *The run on the Rock*, Fifth Report of Session 2007–08.
23. Swiss Federal Chancellery, results of the federal popular initiative *For crisis-safe money* (Vollgeld), 10 June 2018.
24. Basel Committee on Banking Supervision, *Basel III: A global regulatory framework for more resilient banks and banking systems*, revised June 2011; and the LCR and NSFR standards.
25. Cantillon, R., *Essai sur la Nature du Commerce en Général*, written c. 1730, published 1755.
26. Finney, H., forum post on Bitcoin-backed banking, bitcointalk.org, 30 December 2010.
27. The Ratty2Tails Papers, *The Cantillon Effect*, *Quantitative Easing* and *The Claim and the Coin*, Monetary History Series.

THE RATTY2TAILS PAPERS · MONETARY HISTORY SERIES

Written and typeset by Ratty2Tails, September 2026. Set in Source Serif 4, Fraunces, Archivo and EB Garamond. Diagrams hand-authored in SVG. Rendered with WeasyPrint.

Nothing in this paper is financial advice. It is monetary history, an argument about mechanics, and the opinion of a rat.